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Work machine

Abstract

A router includes an illumination part having an LED, and the illumination part is accommodated in an illumination accommodation portion of a holding ring and held by the holding ring. Here, a restriction portion of the holding ring is arranged adjacent to a lower side of a positioning rib of a stator and restricts downward movement of the stator. Thus, the illumination part can be held using the holding ring that restricts the downward movement of the stator. Accordingly, the illumination part can be mounted in the router using a member that holds the illumination part and a member that restricts the downward movement of the stator as a common member. In this manner, the periphery of a tip tool can be illuminated by the illumination part, while preventing an increase in the number of components.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a 371 application of the International PCT application serial no. PCT/JP2020/048842, filed on Dec. 25, 2020, which claims the priority benefits of Japan Patent Application No. 2020-014326, filed on Jan. 31, 2020, and Japan Patent Application No. 2020-164779, filed on Sep. 30, 2020. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

(2) The invention relates to a work machine.

RELATED ART

(3) The following Patent Literature 1 discloses a router (work machine) that performs a cutting operation on a material to be cut through rotation of a tip tool driven by a motor. Specifically, the operator grasps a handle of the router and pushes the router downward, thereby bringing the tip tool into contact with the material to be cut and performing the cutting operation on the material to be cut. In addition, there are routers provided with an illumination part that illuminates the periphery of the tip tool in order to improve the workability during cutting.

CITATION LIST

Patent Literature

(4) [Patent Literature 1] Japanese Patent Laid-Open No. 2006-326933

SUMMARY OF INVENTION

Technical Problem

(5) Here, when the router described in Patent Literature 1 is equipped with an illumination part, it is necessary to separately provide a member for holding the illumination part. Therefore, the number of components may increase. Accordingly, when the illumination part is mounted on the router, it is desirable to have a structure that can illuminate the periphery of the tip tool while preventing an increase in the number of components. Further, when the illumination part is mounted, it is desirable to suppress deterioration of the assemblability.

(6) In view of the above, the invention aims to provide a work machine that can illuminate the periphery of a tip tool while preventing an increase in the number of components. Further, the invention aims to provide a work machine that can illuminate the periphery of a tip tool while suppressing deterioration of the assemblability.

Solution to Problem

(7) One or more embodiments of the invention provide a work machine, including: a motor including a rotating shaft, a rotor configured to be integrally rotatable on the rotating shaft, and a stator arranged on a radially outer side of the rotor; a fan provided on the rotating shaft and

arranged on one axial side of the rotating shaft with respect to the rotor and the stator; a housing accommodating the motor; a tip tool attached to one axial end of the rotating shaft; an illumination part including a light source that irradiates emitted light toward a periphery of the tip tool; and a holding member provided on a radially outer side of the fan, and rectifying an air flow generated by the fan and holding the illumination part.

(8) One or more embodiments of the invention provide a work machine, including: a motor including a rotating shaft, a rotor configured to be integrally rotatable on the rotating shaft, and a stator arranged on a radially outer side of the rotor; a housing accommodating the motor; a fan provided on the rotating shaft and arranged on one axial side of the rotating shaft with respect to the rotor and the stator; a tip tool attached to one axial end of the rotating shaft; an illumination part including a light source that irradiates emitted light toward a periphery of the tip tool; and a holding member provided on one axial side of the rotating shaft with respect to the stator, and holding the illumination part and restricting movement of the stator toward one axial side of the rotating shaft.

(9) In one or more embodiments of the invention, the housing includes: a first housing accommodating the motor; and a second housing provided on one axial side of the rotating shaft with respect to the first housing. The holding member is sandwiched by the first housing and the second housing in an axial direction of the rotating shaft.

(10) In one or more embodiments of the invention, an engaging portion engaged with the holding member is formed in the second housing, and rotation of the holding member in a circumferential direction of the rotating shaft is restricted by the engaging portion.

(11) In one or more embodiments of the invention, the holding member includes: a base portion arranged between the rotor and the stator, and the fan; and an extension portion extending from the base portion to one axial side of the rotating shaft. The illumination part is held at a tip of the extension portion.

(12) In one or more embodiments of the invention, the holding member includes: a base portion arranged between the rotor and the stator, and the fan; and an extension portion extending from the base portion to one axial side of the rotating shaft. The illumination part is held at a base end portion of the extension portion. The extension portion transmits light emitted by the light source, and the light is emitted from a tip of the extension portion toward the periphery of the tip tool.

(13) In one or more embodiments of the invention, a groove that is opened to the other axial side of the rotating shaft is formed in the base portion, and wiring connected to the illumination part is arranged in the groove.

(14) In one or more embodiments of the invention, the holding member holds a pair of the illumination parts, and wiring connecting the pair of the illumination parts is arranged in the groove.

(15) In one or more embodiments of the invention, a split housing is provided on the other axial side of the rotating shaft with respect to the housing, and the split housing is configured to be splittable in a direction orthogonal to the axial direction of the rotating shaft.

(16) In one or more embodiments of the invention, a controller that controls turning on and off of the illumination part is provided in the split housing.

(17) In one or more embodiments of the invention, a battery that supplies electric power to the illumination part is detachably provided in the split housing.

(18) One or more embodiments of the invention provide a work machine, including: a motor including a rotating shaft, a rotor configured to be integrally rotatable on the rotating shaft, and a stator arranged on a radially outer side of the rotor; a fan provided on the rotating shaft and arranged on one axial side of the rotating shaft with respect to the rotor and the stator; a housing accommodating the motor; a tip tool attached to one axial end of the rotating shaft; an illumination part including a light source that irradiates emitted light toward a periphery of the tip tool; and a holding member holding the illumination part. The holding member includes: a base portion arranged between the rotor and the stator, and the fan; and an extension portion extending from the

base portion to one axial side of the rotating shaft and holding the illumination part, and transmitting light emitted by the light source and irradiating the light from a tip toward the periphery of the tip tool. The tip of the extension portion is arranged on one axial side of the rotating shaft with respect to the fan.

Effects of Invention

(19) The work machine having the above configuration can illuminate the periphery of the tip tool while preventing an increase in the number of components.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a front view showing the router according to the present embodiment as viewed from the front side.
- (2) FIG. 2 is a side view of the router shown in FIG. 1 as viewed from the right side.
- (3) FIG. 3 is an exploded perspective view of the router main body of the router shown in FIG. 1 as viewed from the front diagonally to the right.
- (4) FIG. 4 is a cross-sectional view of the router shown in FIG. 1 as viewed from the lower side of the router main body (cross-sectional view taken along the line 4-4 of FIG. 1).
- (5) FIG. 5 is a cross-sectional view showing the inside of the router main body shown in FIG. 4 as viewed from the front side (cross-sectional view taken along the line 5-5 of FIG. 4).
- (6) FIG. 6 is a cross-sectional view showing the inside of the router main body shown in FIG. 4 at the position of the illumination hole as viewed from the front side (cross-sectional view taken along the line 6-6 of FIG. 4).
- (7) FIG. 7 is a plan cross-sectional view showing a state where the motor shown in FIG. 5 is accommodated in the motor accommodation portion at the position of the lower end of the motor as viewed from the upper side (cross-sectional view taken along the line 7-7 of FIG. 5).
- (8) FIG. 8 is a plan cross-sectional view showing a state where the holding ring shown in FIG. 5 is accommodated in the end bracket at the position of the upper end of the holding ring as viewed from the upper side (cross-sectional view taken along the line 8-8 of FIG. 5).
- (9) (A) of FIG. 9 is an enlarged perspective view showing the holding ring shown in FIG. 3, and (B) of FIG. 9 is a front view of the holding ring of (A) as viewed from the front side.
- (10) FIG. 10 is an enlarged plan view showing a state where the illumination part is held in the holding ring shown in FIG. 8 as viewed from the upper side.
- (11) (A) of FIG. 11 is a cross-sectional view showing the ring base of the holding ring shown in FIG. 10 at the position of the reinforcing rib (cross-sectional view taken along the line 11A-11A of FIG. 10), and (B) of FIG. 11 is a cross-sectional view showing the holding ring shown in FIG. 10 around the holding pillar (cross-sectional view taken along the line 11B-11B of FIG. 10). (C) of FIG. 11 is a cross-sectional view showing the vertical intermediate portion of the holding pillar shown in (B) as viewed from the upper side (cross-sectional view taken along the line 11C-11C of (B) of FIG. 11), and (D) of FIG. 11 is a cross-sectional view showing the lower end of the holding pillar shown in (B) as viewed from the upper side (cross-sectional view taken along the line 11D-11D of (B) of FIG. 11).
- (12) (A) of FIG. 12 is a cross-sectional view showing a modified example of the holding ring when the fan shown in FIG. 6 is a centrifugal fan as viewed from the front side, and (B) of FIG. 12 is a cross-sectional view showing the fan and the holding ring of (A) as viewed from the front diagonally to the right.
- (13) FIG. 13 is a cross-sectional view corresponding to (B) of FIG. 11 and showing a modified example of the holding ring when the vertical position of the illumination part shown in (B) of FIG. 11 is changed.

DESCRIPTION OF EMBODIMENTS

(14) Hereinafter, a router **10** serving as a “work machine” according to the present embodiment will be described with reference to the drawings. The arrows UP, FR, and RH appropriately shown in the drawings indicate the upper side, the front side, and the right side of the router **10**, respectively. In the following description, when the up-down, front-rear, and left-right directions are used, they indicate the up-down direction, front-rear direction, and left-right direction of the router **10** unless otherwise specified.

(15) The router **10** is configured as a tool that is placed on the upper side of a material to be cut for cutting the material to be cut. As shown in FIG. 1 and FIG. 2, the router **10** includes a base **20**, a router main body **30**, a pair of illumination parts **70L** and **70R** (see FIG. 8), a battery holder **80** as a “split housing,” a battery **85** as a “cell,” and a controller **90** (see FIG. 5). Hereinafter, each configuration of the router **10** will be described.

(16) (Regarding the base **20**) As shown in FIG. 1 to FIG. 3, the base **20** constitutes the lower end of the router **10** and is formed in a substantially circular plate shape with the up-down direction as the plate thickness direction. An insertion portion **20A** is formed through the substantially central portion of the base **20** in the up-down direction.

(17) A pair of columns **21** are provided at both ends of the base **20** in the left-right direction, and the column **21** is formed in a substantially cylindrical shape with the up-down direction as the axial direction. The lower end of the column **21** is fixed to the base **20**, and the column **21** extends upward from the base **20**.

(18) The base **20** is provided with a bolt **22** extending in the up-down direction on the rear side of the column **21** on the left side. The lower end of the bolt **22** is fixed to the base **20**, and the bolt **22** extends upward from the base **20**. Further, a nut **23** (see FIG. 3) is screwed to the upper end of the bolt **22**.

(19) (Regarding the router main body **30**) As shown in FIG. 1 to FIG. 8, the router main body **30** includes a housing **31** that constitutes the outer shell of the router main body **30**, a motor **40** accommodated inside the housing **31**, a holding ring **60** as a “holding member,” and a pair of left and right handles **50** attached to the housing **31**. Further, the housing **31** includes an end bracket **32** as a “second housing” constituting the lower portion of the housing **31**, and a motor housing **33** as a “first housing” constituting the upper portion of the housing **31**. Hereinafter, each configuration of the router main body **30** will be described.

(20) <Regarding the end bracket **32**> The end bracket **32** is made of a metal material. The end bracket **32** is formed in a substantially rectangular parallelepiped box shape that is opened upward with the left-right direction as the longitudinal direction. The end bracket **32** has a single piece structure and cannot be split. A pair of connecting cylinder portions **32A** are formed at both end side portions of the end bracket **32** in the left-right direction. The connecting cylinder portion **32A** is formed in a substantially cylindrical shape with the up-down direction as the axial direction, and protrudes upward and downward from the bottom wall of the end bracket **32**. Then, the column **21** described above is inserted into the connecting cylinder portion **32A** so as to be relatively movable in the up-down direction.

(21) In the central portion of the end bracket **32** in the left-right direction, a fan accommodation portion **32B** for accommodating a fan **47**, which will be described later, is formed between the pair of connecting cylinder portions **32A**. The fan accommodation portion **32B** is formed in a substantially cylindrical shape and protrudes upward from the bottom wall of the end bracket **32**. Further, the end bracket **32** is formed with a substantially cylindrical support cylinder portion **32C** on the radially inner side of the fan accommodation portion **32B**. The support cylinder portion **32C** is arranged coaxially with the fan accommodation portion **32B**, and the inside of the support cylinder portion **32C** penetrates in the up-down direction. A plurality of relief grooves **32D** (see FIG. 5) opened downward are formed in the support cylinder portion **32C**, and the relief grooves **32D** are arranged side by side along the circumferential direction of the support cylinder portion

32C. Further, a bearing 34 for supporting the rotating shaft 41 of the motor 40, which will be described later, is provided on the radially inner side the support cylinder portion 32C.

(22) In addition, on the bottom wall of the end bracket 32, a plurality of exhaust holes 32E (see FIG. 4) are formed between the fan accommodation portion 32B and the support cylinder portion 32C. The plurality of exhaust holes 32E are arranged side by side in the fan accommodation portion 32B along the circumferential direction. Further, on the radially outer side of the fan accommodation portion 32B, a pair of left and right circular illumination holes 32F (see FIG. 4) serving as “engaging portions” are formed through the bottom wall of the end bracket 32 in the up-down direction. One of the illumination holes 32F is arranged on the rear diagonally to the right with respect to the axis AL of the support cylinder portion 32C, the other one of the illumination holes 32F is arranged on the rear diagonally to the left with respect to the axis AL of the support cylinder portion 32C.

(23) Furthermore, handle fixing portions 32G for fixing the handles 50, which will be described later, are formed at both ends of the end bracket 32 in the left-right direction. The handle fixing portion 32G is formed in a substantially columnar shape with the left-right direction as the axial direction, and protrudes outward from the end bracket 32 in the left-right direction. A cutout portion 32H (see the cross section AA of FIG. 4) is formed on the outer peripheral portion of the handle fixing portion 32G on the right side, and the cutout portion 32H is opened rearward and upward as viewed from the right side.

(24) Further, a stopper portion 32J (see FIG. 4) protruding rearward is formed on the outer peripheral portion of the left end of the end bracket 32. The stopper portion 32J is formed in a substantially cylindrical shape with the up-down direction as the axial direction, and the bolt 22 described above is inserted into the stopper portion 32J. Then, the stopper portion 32J is in contact with the nut 23 provided at the upper end of the bolt 22 from below. As a result, the upward movement of the end bracket 32 (router main body 30) is restricted at the position shown in FIG. 1 and FIG. 2 (hereinafter, the position of the router main body 30 is referred to as an “initial position”).

(25) <Regarding the motor housing 33> The motor housing 33 is made of a resin material, and is arranged adjacent to the upper side of the end bracket 32. The motor housing 33 is formed in a substantially rectangular box shape (cylindrical shape) that is opened downward. The motor housing 33 has a single piece structure and cannot be split in the front-rear direction or the left-right direction, for example. Further, the opening of the motor housing 33 is formed in a shape corresponding to the opening of the end bracket 32. Then, the motor housing 33 is fastened and fixed to the end bracket 32 so as to close the opening of the end bracket 32.

(26) As shown in FIG. 5, a pair of left and right connecting shafts 35 are provided inside the motor housing 33 at positions corresponding to the columns 21 described above. The connecting shaft 35 is formed in a substantially columnar shape with the up-down direction as the axial direction, and extends downward from the top wall of the motor housing 33 and is inserted into the column 21 so as to be relatively movable in the up-down direction. Further, an urging spring 36 configured as a compression coil spring is disposed around the connecting shaft 35. The upper end of the urging spring 36 is locked to the top wall of the motor housing 33, the lower end of the urging spring 36 is locked to the upper end of the column 21, and the motor housing 33 (that is, the router main body 30) is urged upward by the urging spring 36. As a result, the router main body 30 is held at the initial position. Then, the router main body 30 is configured to move relatively downward with respect to the base 20 by pushing the router main body 30 downward against the urging force of the urging spring 36.

(27) As shown in FIG. 3, FIG. 5, and FIG. 7, a motor accommodation portion 33A for accommodating the motor 40, which will be described later, is formed in the central portion of the motor housing 33 in the left-right direction. The motor accommodation portion 33A is formed in a substantially bottomed cylindrical shape that is opened downward, and is arranged coaxially with

the fan accommodation portion **32B** and the support cylinder portion **32C** of the end bracket **32**. Four guide ribs **33B** extending in the up-down direction are formed on the side wall of the motor accommodation portion **33A**. The guide ribs **33B** respectively protrude toward the radially inner side and the radially outer side from the side wall of the motor accommodation portion **33A**, and the lower end of the guide rib **33B** protrudes downward with respect to the side wall of the motor accommodation portion **33A**. Furthermore, one guide rib **33B** is arranged on the rear diagonally to the left with respect to the axis **AL** of the motor accommodation portion **33A**, and the four guide ribs **33B** are arranged every 90 degrees in the circumferential direction of the motor accommodation portion **33A**.

(28) A pair of front and rear right rotation restriction ribs **33C** (see FIG. 7) extending in the up-down direction are formed at the right end of the inner peripheral surface of the motor accommodation portion **33A**, and the pair of right rotation restriction ribs **33C** are arranged at a predetermined interval in the front-rear direction. Further, a pair of left rotation restriction ribs **33D** (see FIG. 7) extending in the up-down direction are formed at the left end of the inner peripheral surface of the motor accommodation portion **33A**. In addition, a pair of left and right axial restriction ribs **33E** (see FIG. 5) are formed on the inner peripheral surface of the motor accommodation portion **33A**. The axial restriction rib **33E** extends from the top wall of the motor accommodation portion **33A** to the vertical intermediate portion of the motor accommodation portion **33A**. Then, the right axial restriction rib **33E** is arranged between the pair of right rotation restriction ribs **33C** and is connected to the pair of right rotation restriction ribs **33C**. Further, the left axial restriction rib **33E** is arranged on the front side of the left rotation restriction rib **33D** and is connected to the left rotation restriction rib **33D**.

(29) As shown in FIG. 3, four communication holes **33F** are formed through the top wall of the motor accommodation portion **33A**. The communication holes **33F** are formed in a substantially fan shape in a plan view, and are arranged every 90 degrees in the circumferential direction of the motor accommodation portion **33A**. Further, a communication groove **33G** opened downward is formed at the right end of the lower end of the motor housing **33**, and the inside and outside of the motor housing **33** are communicated with each other by the communication groove **33G**.

(30) <Regarding the motor **40**> As shown in FIG. 3, FIG. 5, and FIG. 7, the motor **40** is accommodated in the motor accommodation portion **33A** of the motor housing **33**. The motor **40** includes the rotating shaft **41** that takes the up-down direction as the axial direction, a substantially cylindrical rotor **42** fixed to the rotating shaft **41**, and a substantially cylindrical stator **43** arranged on the radially outer side of the rotor **42**.

(31) The rotating shaft **41** is arranged coaxially with the motor accommodation portion **33A**, and the lower end (the end on one axial side) of the rotating shaft **41** protrudes downward from the motor housing **33** and passes through the support cylinder portion **32C** of the end bracket **32**. That is, the axis of the rotating shaft **41** coincides with the axis **AL**. Then, the lower end of the rotating shaft **41** is rotatably supported by the bearing **34** of the end bracket **32**, and the upper end of the rotating shaft **41** is rotatably supported by a bearing **37** fixed to the motor housing **33**. Since the bearing **34** and the bearing **37** are respectively held by the end bracket **32** and the motor housing **33**, which are configured to be not splittable and thus have high rigidity, the rotating shaft **41** can be supported with high accuracy. Further, a tip tool **46** is attached to the lower end of the rotating shaft **41** via a collet chuck **45**. As a result, the tip tool **46** is configured to pass through the insertion portion **20A** of the base **20** by pushing the router main body **30** downward from the initial position.

(32) The stator **43** has a stator holder **44**, and the stator holder **44** is formed in a substantially cylindrical shape. A stator coil (not shown) is wound around the stator holder **44**, and the stator coil is connected to a motor substrate **40A** provided at the upper end of the motor **40**. A pair of left and right positioning ribs **44A** are formed on the outer peripheral portion of the stator holder **44**, and the positioning ribs **44A** extend in the up-down direction.

(33) Then, in a state where the motor **40** is accommodated in the motor accommodation portion

33A, the guide ribs 33B of the motor housing 33 are arranged adjacent to the radially outer side of the stator holder 44 (see FIG. 7). As a result, the stator holder 44 (stator 43) is arranged coaxially with the rotating shaft 41. Further, the right positioning rib 44A is inserted between the pair of right rotation restriction ribs 33C in the motor housing 33, and the left positioning rib 44A is arranged adjacent to the front side of the left rotation restriction rib 33D of the motor housing 33 (see FIG. 7). As a result, the relative rotation of the stator 43 with respect to the motor housing 33 is restricted by the right rotation restriction rib 33C and the left rotation restriction rib 33D, and the positioning ribs 44A. In addition, the positioning rib 44A is arranged adjacent to the lower side of the axial restriction rib 33E of the motor housing 33, and is arranged adjacent to the upper side of the holding ring 60 described later (see FIG. 5). As a result, the vertical movement of the stator 43 is restricted by the axial restriction rib 33E and the holding ring 60.

(34) As shown in FIG. 3, FIG. 5, FIG. 6, and FIG. 8, the fan 47 is provided at the lower end of the rotating shaft 41 to be integrally rotatable on the lower side of the rotor 42 and the stator 43. Specifically, the fan 47 is arranged on the upper side of the support cylinder portion 32C of the end bracket 32 and on the radially inner side of the upper portion of the fan accommodation portion 32B. Further, the upper portion of the fan 47 protrudes upward from the fan accommodation portion 32B. The fan 47 has a plurality of blades 47A. The blades 47A extend in the radial direction of the rotating shaft 41 and are arranged side by side in the rotation direction of the fan 47. Besides, in the present embodiment, the fan 47 is configured as a so-called axial fan, and when the fan 47 rotates, the air (cooling air) flowing in from the intake hole 80B, which will be described later, cools the controller 90 and then flows into the motor housing 33 from the upper opening of the motor housing 33 to cool the motor 40, and the air is rectified to the lower side of the fan 47 and flows out from the exhaust holes 32E to the lower side.

(35) <Regarding the handle 50> As shown in FIG. 1, FIG. 2, and FIG. 5, the handle 50 is formed hollow in a substantially T shape rotated by 90 degrees in the front view. Specifically, the handle 50 includes a substantially cylindrical mounting portion 50A that takes the left-right direction as the axial direction, and a grip portion 50B extending in the up-down direction from the outer side end of the mounting portion 50A in the left-right direction. Further, the handle 50 is split into two in the left-right direction at the portion of the grip portion 50B, and is composed of two members. Specifically, the handle 50 includes a handle main body 51 that constitutes the inner portion of the handle 50 in the left-right direction, and a handle cover 52 that constitutes the outer portion of the handle 50 in the left-right direction. Then, the handle fixing portion 32G of the end bracket 32 is inserted into the mounting portion 50A, and the handle main body 51 is fastened and fixed to the end bracket 32. Furthermore, after the handle main body 51 is fixed to the end bracket 32, the handle cover 52 is fastened and fixed to the handle main body 51 by screws.

(36) On the right handle 50, a switch button 53 is provided on the upper portion so as to be pressed, and a trigger 54 is provided on the rear portion so as to be pulled. Further, inside the right handle 50, a switch circuit 55 is provided to be operated by operating the switch button 53 and the trigger 54, and the switch circuit 55 is fixed to the handle main body 51. The switch circuit 55 has a switch (not shown) to be operated by the switch button 53 and the trigger 54. The switch is electrically connected to the controller 90, which will be described later, and is configured to output an output signal corresponding to the operation states of the switch button 53 and the trigger 54 to the controller 90 described later.

(37) <Regarding the holding ring 60> As shown in FIG. 5, FIG. 6, and FIG. 8 to FIG. 11, the holding ring 60 is made of a transparent resin material that has translucency. The holding ring 60 has a ring base 61 as a “base portion.” The ring base 61 is formed in a substantially ring shape (substantially cylindrical shape) with the up-down direction as the axial direction as a whole, and is formed in a substantially U shape that is opened upward in a cross-sectional view as viewed from the circumferential direction. Specifically, the ring base 61 includes a substantially cylindrical inner cylinder 61A that takes the up-down direction as the axial direction, a cylindrical outer cylinder

61B that is arranged on the radially outer side of the inner cylinder **61A**, and a bottom portion **61C** that connects the lower end of the inner cylinder **61A** and the lower end of the outer cylinder **61B**. As a result, a guide groove **62** serving as a “groove” opened upward is formed inside the ring base **61**.

(38) Then, the ring base **61** is arranged inside the end bracket **32** on the radially outer side of the fan **47**. Specifically, the bottom portion **61C** of the ring base **61** is arranged adjacent to the upper side of the fan accommodation portion **32B** of the end bracket **32**, and the outer cylinder **61B** of the ring base **61** is arranged adjacent to the lower side of a part of the lower end of the motor housing **33**. As a result, the ring base **61** is sandwiched in the up-down direction by the end bracket **32** and the motor housing **33**, and is fixed to both of them. Further, in a state where the ring base **61** is fixed, the upper portion of the blade **47A** of the fan **47** is arranged on the radially inner side of the lower portion of the inner cylinder **61A** of the ring base **61**. As a result, the air in the motor housing **33** is guided by the inner cylinder **61A** of the holding ring **60** and flows to the side of the fan **47** (blade **47A**).

(39) A pair of left and right restriction portions **63** protruding upward are formed on the upper end surface of the inner cylinder **61A**. The restriction portions **63** are arranged at positions corresponding to the positioning ribs **44A** of the stator **43**, and the lower ends of the positioning ribs **44A** are in contact with the upper ends of the restriction portions **63** (see FIG. 5). As a result, the downward movement of the stator **43** is restricted by the holding ring **60**.

(40) Further, four guide recesses **64** are formed at the upper end of the inner cylinder **61A**. The guide recess **64** is formed in a concave shape that is opened upward, and the guide recesses **64** are arranged at positions corresponding to the guide ribs **33B** of the motor housing **33**. Then, the lower end of the guide rib **33B** is inserted into the guide recess **64** (see FIG. 5, FIG. 6, and FIG. 8). As a result, the motor housing **33** is guided by the holding ring **60** when the motor housing **33** is attached to the end bracket **32**. Further, in a state where the guide rib **33B** is inserted into the guide recess **64**, the lower end of the guide rib **33B** extends in the radial direction of the ring base **61** in the upper portion of the guide groove **62** of the holding ring **60** (see FIG. 8).

(41) In addition, the holding ring **60** has a pair of left and right holding pillars **65** as “extension portions” for holding the illumination parts **70L** and **70R**, which will be described later. The holding pillar **65** is formed in a substantially rectangular columnar shape extending in the up-down direction. Further, the holding pillar **65** is arranged at a position corresponding to the illumination hole **32F** of the end bracket **32** described above (see FIG. 8), and is arranged adjacent to the radially outer side of the ring base **61**. Specifically, one of the holding pillars **65** is arranged on the rear diagonally to the right with respect to the axis **AL** of the rotating shaft **41** in a plan view, and the other one of the holding pillars **65** is arranged on the rear diagonally to the left with respect to the axis **AL** of the rotating shaft **41** in a plan view. Then, the upper end of the holding pillar **65** is connected to the lower end of the outer cylinder **61B** of the ring base **61**, and the holding pillar **65** is integrally formed with the ring base **61**. That is, the holding pillar **65** extends downward from the ring base **61**. Specifically, the lower end of the holding pillar **65** is arranged below the fan **47**. In other words, the holding pillar **65** extends in the up-down direction on the radially outer side of the fan **47**.

(42) The holding pillar **65** is inclined at approximately 45 degrees in a plan view, and the rear surface **65A** of the holding pillar **65** is inclined to the rear side toward the right side in a plan view. The illumination accommodation portions **66** (see (B) and (D) of FIG. 11) for accommodating the illumination parts **70L** and **70R**, which will be described later, are formed at the lower ends of the rear surfaces **65A** of the holding pillars **65** (specifically, the portions below the fan **47**), and the illumination accommodation portion **66** is formed in a concave shape that is opened rearward (specifically, in a direction orthogonal to the rear surface **65A**). Further, a pair of wiring grooves **67** (see (B) and (C) of FIG. 11) that are opened rearward are formed on the rear surface **65A** of the holding pillar **65**. The wiring grooves **67** extend in the up-down direction and are arranged side by

side in the width direction of the holding pillars **65**. The upper end of the wiring groove **67** is opened upward, and the lower end of the wiring groove **67** communicates with the illumination accommodation portion **66**.

(43) A lens portion **65B** (an element grasped as an “engaged portion” in a broad sense) is formed at the lower end of the holding pillar **65**. The lens portion **65B** is formed in a substantially columnar shape with the up-down direction as the axial direction, and protrudes downward from the holding pillar **65**. The lens portion **65B** is fitted into the illumination hole **32F** of the end bracket **32**. As a result, the lens portion **65B** and the illumination hole **32F** are engaged with each other around the axis **AL**, and the position of the holding ring **60** with respect to the end bracket **32** is determined. Specifically, the ring base **61** is arranged coaxially with the rotating shaft **41**.

(44) A pair of wiring slits **68L** and **68R** are formed in the outer cylinder **61B** of the ring base **61** at positions corresponding to the pair of left and right holding pillars **65**. The pair of wiring slits **68L** and **68R** are formed through the outer cylinder **61B** in the radial direction, and are opened upward. As a result, the inside and outside of the ring base **61** are communicated with each other by the wiring slits **68L** and **68R**.

(45) In addition, four reinforcing ribs **69** are formed inside the guide groove **62** of the ring base **61**. The reinforcing rib **69** is formed in a plate shape with the circumferential direction of the ring base **61** as the plate thickness direction, and is connected to the inner cylinder **61A**, the bottom portion **61C**, and the outer cylinder **61B**. One reinforcing rib **69** is arranged at the rear end of the ring base **61**, and four reinforcing ribs **69** are arranged every 90 degrees in the circumferential direction of the ring base **61**. A wiring slit **69A** is formed in the reinforcing rib **69**. The wiring slit **69A** is formed through the reinforcing rib **69** in the plate thickness direction, and is opened upward.

(46) (Regarding the illumination parts **70L** and **70R**) As shown in FIG. **8**, FIG. **10**, and FIG. **11**, the illumination parts **70L** and **70R** each have an illumination substrate **71** and an LED **72** as a “light source.”

(47) The illumination substrate **71** is formed in a substantially rectangular plate shape with the up-down direction as the plate thickness direction, and is accommodated in the illumination accommodation portion **66** of the holding ring **60**. The LED **72** is mounted on the lower surface of the illumination substrate **71**, and is accommodated in the illumination accommodation portion **66** together with the illumination substrate **71**. As a result, the LED **72** (illumination part **70L**, **70R**) is arranged below the fan **47**. Then, the LED **72** is configured to irradiate the emitted light downward. Specifically, the light emitted by the LED **72** passes through the lens portion **65B** of the holding ring **60** and irradiates the lower side of the end bracket **32** from the illumination hole **32F** of the end bracket **32**. As a result, the light illuminates the periphery of the tip tool **46**. More specifically, the light emitted by the LED **72** is configured to illuminate the periphery of the tip tool **46** on the rear side of the tip tool **46** and on the outer side in the left-right direction.

(48) Furthermore, in the left illumination part **70L**, two lead wires **73** serving as “wiring” extend from the illumination substrate **71**, and the lead wires **73** are arranged in the wiring groove **67** of the holding pillar **65** on the left side of the holding ring **60**. Further, the lead wire **73** is inserted into the guide groove **62** of the ring base **61** from the wiring slit **68L** on the left side of the holding ring **60**, and is arranged in the rear portion of the guide groove **62** and extends from the right wiring slit **68R** to the outside of the guide groove **62**. Specifically, the lead wire **73** passes through the wiring slit **69A** of the reinforcing rib **69** on the rear side and extends from the right wiring slit **68R** to the outside of the guide groove **62**. In addition, in a state where the lead wire **73** is arranged in the guide groove **62**, the lower end of the guide rib **33B** of the motor housing **33** is arranged on the upper side of the lead wire **73**, and the upward movement of the lead wire **73** is restricted by the guide rib **33B** (see FIG. **8**).

(49) Then, one of the two lead wires **73** is arranged in one of the wiring grooves **67** of the right holding pillar **65**, and is connected to the illumination substrate **71** of the right illumination part **70R**. Further, a lead wire **74** serving as “wiring” extends from the illumination substrate **71** of the

right illumination part **70R**, and the lead wire **74** is arranged in the other one of the wiring grooves **67** in the right holding pillar **65**.

(50) Then, the other one of the lead wires **73** and the lead wire **74** pass through the mounting portion **50A** of the right handle **50** and are connected to the switch circuit **55**. Specifically, the lead wire **73** and the lead wire **74** pass through the communication groove **33G** of the motor housing **33** and the cutout portion **32H** of the handle fixing portion **32G** in the end bracket **32** and are connected to the switch circuit **55**.

(51) (Regarding the battery holder **80**) As shown in FIG. **1** to FIG. **3**, the battery holder **80** is provided on the upper side of the motor housing **33**, and is formed in a substantially box shape that is opened downward. The battery holder **80** is split into two in the front-rear direction. That is, the battery holder **80** includes a front holder **81** that constitutes the front portion of the battery holder **80**, and a rear holder **82**. The front holder **81** and the rear holder **82** are fastened and fixed to the motor housing **33**.

(52) The battery holder **80** is formed with a battery mounting portion **80A** for mounting the battery **85** described later, and the battery mounting portion **80A** is formed in a concave shape that is opened upward and leftward. Further, the intake hole **80B** that functions as an inlet for the air flow generated by the rotation of the fan **47** is formed near the vertical center of the battery holder **80**. In detail, the intake hole **80B** is formed on the front side wall of the front holder **81** and the rear side wall of the rear holder **82**. In addition, the battery holder **80** is provided with a connector **83** (see FIG. **3**), and the upper portion of the connector **83** is exposed inside the battery mounting portion **80A** so as to be connected to the battery **85** described later.

(53) (Regarding the battery **85**) The battery **85** is formed in a substantially rectangular parallelepiped shape. Then, the battery **85** is attached to the battery mounting portion **80A** of the battery holder **80** from the left side. The battery **85** has a connector (not shown), and in a state where the battery **85** is attached to the battery mounting portion **80A**, the connector is connected to the connector **83**, and electric power is supplied to the controller **90** described later. Furthermore, the battery **85** has a pair of lock members **86**, and the lock members **86** are provided on the front and rear side portions of the battery **85**. Then, in a state where the battery **85** is attached to the battery mounting portion **80A**, the lock members **86** engage with the battery holder **80** to maintain the attached state of the battery **85**.

(54) (Regarding the controller **90**) As shown in FIG. **3**, the controller **90** is accommodated inside the battery holder **80** and is fixed to the battery holder **80**. The controller **90** has a control board (not shown), and the motor **40** and the connector **83** described above are electrically connected to the control board. Specifically, a lead wire (not shown) extending from the motor substrate **40A** passes through the communication hole **33F** of the motor housing **33** and is connected to the control board. Further, a lead wire (not shown) extending from the control board is connected to the switch circuit **55**, and the controller **90** and the switch circuit **55** are electrically connected. Specifically, the lead wire is arranged in the motor housing **33** on the radially outer side of the motor accommodation portion **33A**, and passes through the communication groove **33G** of the motor housing **33** and the cutout portion **32H** of the handle fixing portion **32G** in the end bracket **32** to be connected to the switch circuit **55**. As a result, the controller **90** and the illumination parts **70L** and **70R** are electrically connected. Then, the controller **90** is configured to turn on or off the LED **72** according to the on/off operation of the switch button **53**. The controller **90** is also configured to control the operation for the motor **40** when the trigger **54** is operated with the switch button **53** on.

(55) (Function/effect) Next, the function/effect of the router **10** of the present embodiment will be described.

(56) The router **10** configured as described above includes the illumination parts **70L** and **70R** that have the LEDs **72**. The illumination parts **70L** and **70R** are accommodated in the illumination accommodation portions **66** of the holding ring **60** and are held by the holding ring **60**. Then, the

operator grips the grip portion **50B** of the handle **50** and turns on the switch button **53**, so that the electric power of the battery **85** is supplied to the LED **72** by the controller **90** for the LED **72** to emit light. The light emitted by the LED **72** passes through the lens portion **65B** of the holding ring **60** and goes downward from the illumination hole **32F** of the end bracket **32** to illuminate the periphery of the tip tool **46**. Specifically, the portion on the outer side in the left-right direction and the rear side of the tip tool **46** is illuminated.

(57) Then, when the operator pulls the trigger **54** in this state, the rotating shaft **41** of the motor **40** rotates together with the tip tool **46** under the control of the controller **90**. Further, when the operator pushes the handle **50** downward in this state, the router main body **30** moves relatively downward with respect to the base **20**. As a result, the tip tool **46** comes into contact with the material to be cut and cuts the material to be cut. Further, the operator cuts the material to be cut while moving the router **10** rearward. As a result, the portion of the router **10** on the moving direction side with respect to the tip tool **46** is illuminated by the LED **72**. As a result, it is possible to improve the visibility around the tip tool **46** during cutting. Accordingly, the workability for the operator can be improved.

(58) Here, the holding ring **60** for holding the illumination parts **70L** and **70R** is provided on the lower side of the stator **43** of the motor **40**. The restriction portion **63** of the holding ring **60** is arranged adjacent to the lower side of the positioning rib **44A** of the stator **43** to restrict the downward movement of the stator **43**. That is, the holding ring **60** has two functions, i.e., a function of holding the illumination parts **70L** and **70R** and a function of restricting the downward movement of the stator **43**. In other words, the illumination parts **70L** and **70R** can be held by utilizing the holding ring **60** that restricts the downward movement of the stator **43**. Therefore, the illumination parts **70L** and **70R** can be mounted on the router **10** by using the member for holding the illumination parts **70L** and **70R** and the member for restricting the downward movement of the stator **43** as a common member. According to the above, it is possible to illuminate the periphery of the tip tool **46** with the illumination parts **70L** and **70R** while preventing an increase in the number of components.

(59) Furthermore, the upper portion of the fan **47** is arranged on the radially inner side of the lower portion of the ring base **61** in the holding ring **60**. As a result, the air on the upper side of the fan **47** can be guided by the inner cylinder **61A** of the ring base **61** and guided to the side of the blades **47A** of the fan **47**. Accordingly, the air in the motor housing **33** can be efficiently guided to the fan **47** by utilizing the holding ring **60** that restricts the downward movement of the stator **43**.

(60) Besides, in the holding ring **60**, the bottom portion **61C** of the ring base **61** is arranged adjacent to the upper side of the fan accommodation portion **32B** of the end bracket **32**, and the outer cylinder **61B** of the ring base **61** is arranged adjacent to the lower side of a part of the lower end of the motor housing **33**. As a result, the ring base **61** is sandwiched in the up-down direction by the end bracket **32** and the motor housing **33**, and is fixed to both of them. Therefore, no additional component is required for fixing, and the holding ring **60** can be fixed to the end bracket **32** and the motor housing **33** with a simple configuration while maintaining the single piece structure of the end bracket **32** and the motor housing **33**.

(61) Further, a pair of lens portions **65B** are formed in the holding ring **60**, and the pair of lens portions **65B** are fitted into the illumination holes **32F** of the end bracket **32**. As a result, the holding ring **60** and the end bracket **32** can be engaged with each other around the axis AL of the rotating shaft **41** to determine the position of the holding ring **60** with respect to the end bracket **32**. In other words, it is possible to determine the position of the holding ring **60** with respect to the end bracket **32** while restricting the rotation of the rotating shaft **41** around the axis AL in the holding ring **60**. As a result, the motor housing **33** can be assembled to the end bracket **32** while the holding ring **60** is temporarily fixed to the end bracket **32**, and the holding ring **60** can be fixed by the end bracket **32** and the motor housing **33**. Accordingly, the assemblability of the motor housing **33** to the end bracket **32** can be improved.

(62) Furthermore, the holding ring **60** includes the ring base **61** that is arranged between the motor **40** (rotor **42** and stator **43**) and the fan **47**, and the holding pillars **65** that extend downward from the ring base **61**. Then, the illumination parts **70L** and **70R** are accommodated in the illumination accommodation portions **66** formed at the lower ends of the holding pillars **65**. As a result, while the downward movement of the stator **43** is restricted by the ring base **61**, the holding pillars **65** can hold the LEDs **72** of the illumination parts **70L** and **70R** at positions close to the tip tool **46**.

Accordingly, the illumination effect of the illumination parts **70L** and **70R** for the periphery of the tip tool **46** can be enhanced. In particular, the arrangement of the illumination substrate **71** and the LED **72** in the bottom portion of the end bracket **32** that has a box-shaped (cylindrical) single piece structure makes it difficult to assemble, but by forming the holding pillars **65** extending downward and arranging the holding ring **60**, two illumination arrangements can be easily made.

(63) Besides, the illumination parts **70L** and **70R** (illumination accommodation portions **66**) are arranged below the fan **47**, and the wiring groove **67** extending in the up-down direction is formed in the holding pillar **65** of the holding ring **60**. Then, the lead wires **73** and **74** extending from the illumination parts **70L** and **70R** are arranged in the wiring grooves **67**. That is, on the radially outer side of the fan **47**, the holding pillars **65** extend in the up-down direction, and the lead wires **73** and **74** are held by the holding pillars **65**. As a result, it is possible to arrange the illumination parts **70L** and **70R** at positions close to the tip tool **46** while preventing the lead wires **73** and **74** extending from the illumination parts **70L** and **70R** from moving to the side of the fan **47**.

(64) Further, the ring base **61** of the holding ring **60** is formed in a ring shape, and the ring base **61** has the guide groove **62** that extends over the entire circumference in the circumferential direction. Then, the lead wire **73** extending from the illumination part **70L** is arranged in the guide groove **62**. As a result, it is possible to arrange the lead wire **73** around the fan **47** while the ring base **61** (guide groove **62**) prevents the lead wire **73** for supplying electric power to the illumination part **70L** from moving to the side of the fan **47**.

(65) In addition, the holding ring **60** is provided with a pair of illumination parts **70L** and **70R**, and the illumination parts **70L** and **70R** are connected by the lead wire **73** arranged in the guide groove **62**. Therefore, it is possible to connect the pair of illumination parts **70L** and **70R** arranged around the fan **47** with the lead wire **73** while the ring base **61** (guide groove **62**) prevents the lead wire **73** from moving to the side of the fan **47**. In other words, the lead wire **73** can be prevented from being displaced from a desired position during assembly or driving.

(66) Furthermore, the ring base **61** of the holding ring **60** is formed with the reinforcing ribs **69** inside the guide groove **62**, and the reinforcing ribs **69** are connected to the inner cylinder **61A**, the outer cylinder **61B**, and the bottom portion **61C** of the ring base **61**. Further, the reinforcing rib **69** is formed with the wiring slit **69A** that penetrates in the circumferential direction of the guide groove **62**, and the lead wire **73** arranged in the guide groove **62** is inserted into the wiring slit **69A**. As a result, the lead wire **73** in the guide groove **62** can be held while the ring base **61** is reinforced by the reinforcing rib **69**.

(67) Moreover, the guide recesses **64** are formed in the inner cylinder **61A** of the holding ring **60**. Then, the guide rib **33B** of the motor housing **33** is inserted into the guide recess **64** and is arranged in the upper portion of the guide groove **62** of the ring base **61**. Therefore, the guide rib **33B** of the motor housing **33** can restrict the upward movement of the lead wire **73** arranged in the guide groove **62**. As a result, the guide rib **33B** for determining the position of the stator **43** with respect to the motor housing **33** can be utilized to prevent the lead wire **73** from coming out of the opening of the guide groove **62**. Accordingly, it is possible to maintain a better arrangement state of the lead wire **73** in the guide groove **62**.

(68) In the present embodiment, the fan **47** is configured as a so-called axial fan, but the fan **47** may be configured as a so-called centrifugal fan. In this case, the air flow generated by the fan **47** may be rectified in the holding ring **60**. Hereinafter, the configuration in which the fan **47** is a centrifugal fan will be described with reference to (A) and (B) of FIG. 12.

(69) As shown in these figures, when the fan **47** is configured as a centrifugal fan, the blades **47A** of the fan **47** are arranged on the radially inner side of the lower end of the inner cylinder **61A** in the holding ring **60**. Further, a guide inclined portion **60A** is formed at the lower end of the inner cylinder **61A**, and the guide inclined portion **60A** is inclined downward as it goes toward the radially outer side of the ring base **61** in a cross-sectional view as viewed from the circumferential direction of the ring base **61**. Furthermore, the bottom portion **61C** of the ring base **61** is formed with a guide piece **60B** on the radially outer side of the ring base **61** with respect to the guide inclined portion **60A**, and the guide piece **60B** is formed in a substantially cylindrical shape that extends downward from the bottom portion **61C**.

(70) Since the fan **47** is configured as a centrifugal fan, the fan **47** generates an air flow AC that flows from the center side of the fan **47** to the radially outer side. At this time, since the opening on the upper side (downstream side of the air flow AC) of the holding ring **60** has a diameter smaller than the outer diameter of the fan **47** (blades **47A**), air can be efficiently collected on the side of the rotating shaft **41**, and the flow of the air flow AC flowing from the center side to the radially outer side can be smoothly guided. Further, the air flow AC flowing to the radially outer side of the fan **47** is rectified by the guide inclined portion **60A**, and the direction of the air flow AC is changed downward. Besides, the air flow AC changed to flow downward by the guide inclined portion **60A** is guided by the guide piece **60B** and flows downward from the holding ring **60**. Then, the air flow AC flowing downward from the holding ring **60** is discharged downward from the exhaust holes **32E** of the end bracket **32**. As described above, the air flow AC generated by the fan **47** can be rectified by the holding ring **60** and discharged from the exhaust holes **32E** of the end bracket **32**. Accordingly, the holding ring **60** that holds the illumination parts **70L** and **70R** and restricts the downward movement of the stator **43** can be utilized to rectify the air flow AC generated by the fan **47**. Since it is not necessary to separately provide a member for rectifying the air flow AC, it is possible to more effectively prevent an increase in the number of components.

(71) Furthermore, when the fan **47** is configured as a centrifugal fan and the air flow AC generated by the fan **47** is rectified by the holding ring **60**, the restriction portion **63** in the holding ring **60** may be omitted, and the downward movement of the stator **43** may be restricted by another member. For example, the downward movement of the stator **43** may be restricted by the end bracket **32**.

(72) Additionally, in the present embodiment, the illumination accommodation portion **66** is formed at the lower end of the holding pillar **65** in the holding ring **60**, but the vertical position of the illumination accommodation portion **66** can be arbitrarily changed. That is, the positions of the illumination parts **70L** and **70R** in the up-down direction can be arbitrarily changed.

(73) For example, as shown in FIG. **13**, the illumination part **70R** (illumination part **70L**) may be configured to be held at the upper end of the holding pillar **65**. In this case, the illumination accommodation portion **66** is formed at the upper end of the holding pillar **65**, and is formed in a concave shape that is opened upward. Further, the wiring groove **67** in the holding pillar **65** is omitted. Then, the illumination part **70R** (illumination part **70L**) is accommodated in the illumination accommodation portion **66** from above to be held by the holding pillar **65**. Besides, in this case, the light emitted by the LED **72** of the illumination part **70R** (illumination part **70L**) (see the arrows indicated by the two-dot chain lines in FIG. **13**) passes through the portion of the holding pillar **65** below the illumination accommodation portion **66** and irradiates the periphery of the tip tool **46**. That is, the holding pillar **65** including the lens portion **65B** functions as a light guide portion that guides the light emitted by the LED **72** downward, and can guide the light to the lower side of the fan **47**. Then, the light can be irradiated downward from the illumination hole **32F** of the end bracket **32** to illuminate the periphery of the tip tool **46**. Further, in this case, it is not necessary to arrange the lead wires **73** and **74** in the wiring groove **67** of the holding pillar **65**. Accordingly, it is possible to reduce the assembly man-hours when assembling the illumination parts **70L** and **70R** to the holding ring **60**. Furthermore, it is possible to omit the wiring area of the

lead wire **73** extending downward from the initial position (position of the wiring slit **68R**) at which the lead wire **73** is attached to the holding ring **60**. Therefore, it is possible to achieve the same effect as the guide groove **62**, which is to prevent fluttering, that is, displacement, of the lead wire **73** in this area.

(74) Further, in the present embodiment, a pair of illumination parts **70L** and **70R** are held by the holding ring **60**, but the number of illumination members held by the holding ring **60** may be arbitrarily set. That is, the number of illumination members held by the holding ring **60** may be one or three or more. Besides, although a router has been described as an embodiment of the invention, the invention can be applied to various other work machines. In particular, the invention is applicable to a work machine in which the fan is located between the tip tool and the motor, and is also applicable to a work machine such as a disc grinder or a jigsaw.

Claims

1. A work machine, comprising: a motor comprising a rotating shaft extending along an up-down direction, a rotor configured to be integrally rotatable on the rotating shaft, and a stator arranged on a radially outer side of the rotor; a fan provided on the rotating shaft and arranged below the stator; a motor housing accommodating the motor; a tip tool attached to a lower end of the rotating shaft; a pair of columns, extending along the up-down direction; an end bracket, configured separately from the motor housing the end bracket holding a bearing for supporting the rotating shaft of the motor and accommodating at least a part of the pair of columns; a right handle and a left handle, fixed to a right side and a left side of the end bracket respectively; a switch circuit unit, configured to be operated by an operator to drive the motor the switch circuit unit disposed on one of the right handle or the left handle; a base, fixed to the pair of columns; an illumination part comprising a light source that irradiates emitted light toward a periphery of the tip tool; and a holding portion extending from a high position above the fan to a low position below the fan and holding the illumination part, wherein the illumination part is positioned offset from a line that connects axes of the pair of columns when viewed along the up-down direction.
2. The work machine according to claim 1, wherein the end bracket is provided on a lower side of the motor housing, and wherein the holding portion is sandwiched by the motor housing and the end bracket in the up-down direction.
3. The work machine according to claim 1, wherein an engaging portion engaged with the holding portion is formed in the end bracket, and rotation of the holding portion in a circumferential direction of the rotating shaft is restricted by the engaging portion.
4. The work machine according to claim 1, wherein the illumination part is held at a lower end of the holding portion.
5. The work machine according to claim 1, wherein the illumination part is held at an upper end portion of the holding portion, the holding portion transmits light emitted by the light source, and the light is emitted from a lower end of the holding portion toward the periphery of the tip tool.
6. The work machine according to claim 1, wherein a groove that is opened upward is formed in the holding portion, and a wiring for supplying electric power to the illumination part is connected to the illumination part and arranged in the groove.
7. The work machine according to claim 1, wherein the holding portion holds a pair of illumination parts, and a wiring connecting the pair of illumination parts is arranged in a groove of the holding portion.
8. The work machine according to claim 1, wherein a split housing is provided on an upper side of the motor housing, and the split housing is configured to be splittable in a direction orthogonal to the up-down direction.
9. The work machine according to claim 8, wherein a controller that controls turning on and off of the illumination part is provided in the split housing.

10. The work machine according to claim 8, wherein a battery that supplies electric power to the illumination part is detachably provided in the split housing.
11. The work machine according to claim 1, wherein when viewed along the up-down direction, the illumination part is proximate to one of the columns.
12. The work machine according to claim 1, wherein the holding portion extends from an inside of the motor housing to an inside of the end bracket.
13. The work machine according to claim 1, further comprising a controller that controls turning on and off of the illumination part, wherein the illumination part includes a first illumination part and a second illumination part, wherein the holding portion has a first holding part holding the first illumination part and a second holding part holding the second illumination part, wherein the first illumination part is electrically connected to the controller by a first lead wire, and the second illumination part is electrically connected to the controller by a second wire.
14. The work machine according to claim 1, wherein the illumination part includes a substrate and a LED mounted on the substrate, the holding portion holding the substrate so that the substrate intersects the up-down direction.
15. The work machine according to claim 14, further comprising a controller that controls turning on and off of the illumination part, wherein the holding portion has a slit, the illumination part is electrically connected to the controller by a lead wire passing through the slit.
16. The work machine according to claim 15, wherein movement of the lead wire is restricted by the slit.
17. A work machine, comprising: a motor comprising a rotating shaft extending along an up-down direction, a rotor configured to be integrally rotatable on the rotating shaft, and a stator arranged on a radially outer side of the rotor; a fan provided on the rotating shaft and arranged below the stator; a motor housing accommodating the motor; a tip tool attached to a lower end of the rotating shaft; a pair of columns, having a cylindrical shape extending along the up-down direction; an end bracket, configured separately from the motor housing and connected to a lower portion of the motor housing the end bracket holding a bearing for supporting the rotating shaft of the motor and accommodating at least a part of the pair of columns; a right handle and a left handle, fixed to a right side and a left side of the end bracket respectively; a switch circuit unit, configured to be operated by an operator to drive the motor the switch circuit unit disposed on one of the right handle or the left handle; a base, fixed to the pair of columns; an illumination part comprising a light source that irradiates emitted light toward a periphery of the tip tool; and a holding portion holding the illumination part, wherein a wall portion extending along the up-down direction is provided at a radially outer side of the fan in the end bracket, wherein the end bracket has an illumination hole penetrating the end bracket in the up-down direction, the illumination hole being provided at a radially outer side of the wall portion, wherein the holding portion extends from a high position above the fan to a low position below the fan, and irradiates the light from a lower end of the holding portion toward the periphery of the tip tool, and wherein the holding portion is provided at a radially outer side of the wall portion, and the lower end of the holding portion is arranged below the fan and is inserted in the illumination hole.
18. A work machine, comprising: a motor comprising a rotating shaft extending along an up-down direction, a rotor configured to be integrally rotatable on the rotating shaft, and a stator arranged on a radially outer side of the rotor; a centrifugal fan provided on the rotating shaft and arranged below the stator; a motor housing accommodating the motor; a tip tool attached to a lower end of the rotating shaft; a pair of columns, having a cylindrical shape extending along the up-down direction; an end bracket, configured separately from the motor housing and connected to a lower portion of the motor housing the end bracket holding a bearing for supporting the rotating shaft of the motor and accommodating at least a part of the pair of columns; a right handle and a left handle, fixed to a right side and a left side of the end bracket respectively; a switch circuit unit, configured to be operated by an operator to drive the motor the switch circuit unit disposed on one

of the right handle or the left handle; a base, fixed to the pair of columns; an illumination part comprising a light source that irradiates emitted light toward a periphery of the tip tool; and a holding member holding the illumination part, wherein the holding member comprises: a first portion extending along the up-down direction, a second portion connected to the first portion, and an extension portion connected to the first portion, the extension portion extending downward and holding the illumination part, wherein the second portion has a through hole opening along the up-down direction, the through hole opening being smaller than an outer diameter of the centrifugal fan, and wherein an air flow generated by the fan is guided by the first portion and the second portion, and a lower end of the extension portion irradiates the light to the periphery of the tip tool.
